SmartBanana – Ripeness Classification and Quality Recommendation System For BANANANANANANANA

BMDS2133 Image Processing Assignment Documentation

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Introduction

1.1 Problem Background

Bananas are among the most widely cultivated and consumed fruits worldwide due to their high nutritional value, affordability, and year-round availability. They are an important agricultural commodity that contributes significantly to food security, household consumption, and the global fruit trade. According to the Food and Agriculture Organization (FAO), millions of tonnes of bananas are produced annually, making bananas one of the world's most important fruit crops. The commercial value of bananas depends heavily on their external appearance, ripeness, and overall quality, as these characteristics directly influence consumer purchasing decisions, market price, and post-harvest losses (Food and Agriculture Organization [FAO], 2023). Consequently, ensuring accurate and consistent quality assessment has become increasingly important throughout the fruit supply chain, including farms, wholesalers, retailers, supermarkets, and households.

Banana ripeness refers to the physiological maturity stage of a banana during the natural ripening process. As bananas mature, biochemical changes occur within the fruit, including the degradation of chlorophyll, the synthesis of carotenoid pigments, and the conversion of starch into soluble sugars. These processes gradually change the peel colour from green to yellow before eventually developing brown or black spots as senescence begins. The ripeness stage influences not only the fruit's taste, texture, aroma, and nutritional quality but also its remaining storage life and commercial value. Bananas harvested or sold at an inappropriate ripeness stage may experience accelerated spoilage, resulting in unnecessary food waste and financial losses for retailers and consumers (Kader, 2022).

Quality assessment is the process of evaluating whether a fruit satisfies predefined quality standards based on its physical characteristics. For bananas, external quality is primarily determined through visual indicators such as peel colour, surface cleanliness, bruises, black spots, fungal infection, mechanical damage, and other visible defects. Fruits with minimal defects generally receive higher market grades and command better selling prices, whereas bananas with extensive bruising or decay are often discounted or discarded. Since these quality characteristics directly influence consumer acceptance, reliable quality assessment is essential for effective inventory management, pricing strategies, and reducing post-harvest losses (Brosnan & Sun, 2021).

In current industrial and retail practice, banana ripeness and quality are still predominantly assessed through manual visual inspection. Experienced workers estimate the maturity level by observing peel colour and identifying visible defects before assigning a quality grade. Although manual inspection is inexpensive and requires minimal equipment, it presents several limitations. The assessment is highly subjective because different inspectors may interpret colour variations differently depending on their experience and perception. Furthermore, environmental conditions such as illumination intensity, shadow, viewing angle, and surrounding background may influence human judgement, leading to inconsistent grading results. Manual inspection also becomes increasingly inefficient when processing large quantities of bananas, as the procedure is labour-intensive, time-consuming, and susceptible to human fatigue. These limitations reduce grading consistency and operational efficiency, particularly in commercial environments that require rapid inspection of large fruit batches (Chuquimarca et al., 2024).

Image processing refers to the computational analysis and manipulation of digital images to extract meaningful information automatically. Unlike artificial intelligence methods that require extensive model training and large labelled datasets, classical image processing applies deterministic algorithms to analyse visual characteristics such as colour, intensity, shape, texture, and object boundaries. Common image processing operations include image enhancement, colour space transformation, thresholding, morphological processing, contour extraction, and connected component analysis. These techniques enable the automated identification of important fruit characteristics while maintaining relatively low computational complexity, making them suitable for educational applications and practical deployment on standard computing devices (Gonzalez & Woods, 2018).

Image preprocessing is an essential stage in any image processing pipeline because the quality of the input image directly affects the reliability of subsequent analysis. Images captured under different lighting conditions or using different cameras often contain noise, uneven illumination, or low contrast that may reduce segmentation accuracy. Consequently, preprocessing techniques such as image resizing, Gaussian filtering, median filtering, histogram equalisation, and contrast stretching are commonly applied to improve image quality before feature extraction. These operations reduce image noise, enhance colour contrast, and standardise image dimensions, thereby increasing the robustness of later segmentation and analysis processes (Sonka et al., 2015).

Following preprocessing, banana segmentation is performed to isolate the fruit from its surrounding background. Accurate segmentation ensures that only banana pixels contribute to subsequent colour analysis and defect detection. Colour space conversion from RGB to HSV is widely employed because the HSV representation separates colour information from illumination, allowing peel colour to be analysed more consistently under varying lighting conditions. After colour conversion, HSV thresholding and binary masking are applied to distinguish banana regions from the background. Morphological opening and closing remove small artefacts and fill holes within the segmented object, while contour detection identifies the fruit boundary. Selecting the largest contour enables the system to isolate the primary banana object accurately, thereby improving the reliability of subsequent feature extraction and quality analysis.

Ripeness assessment within the proposed system is based on quantitative colour analysis rather than subjective human observation. Instead of relying solely on visual judgement, the system measures the proportions of green, yellow, and brown pixels across the banana peel using HSV colour analysis. Hue histograms and colour percentage calculations provide objective numerical indicators representing the fruit's maturity level. By comparing these colour distributions against predefined ripeness thresholds, the system classifies the banana into appropriate ripeness categories and provides consistent assessments independent of individual user experience. Such rule-based colour analysis eliminates the variability commonly associated with manual inspection while maintaining transparent decision-making processes.

Besides ripeness assessment, surface defects provide another important indicator of banana quality. Mechanical bruises, black spots, fungal infection, and surface discolouration frequently reduce commercial value despite the fruit remaining edible. Consequently, objective defect detection has become increasingly important in modern fruit quality assessment systems. The proposed system identifies dark surface defects through thresholding techniques, morphological filtering, and connected component analysis. After detecting all defect regions, the total defect area is calculated as

a percentage of the segmented banana surface. This quantitative measurement enables the system to evaluate fruit quality objectively instead of relying solely on subjective visual estimation. Based on predefined grading rules that combine ripeness information with defect percentage, the system assigns an appropriate quality grade for each analysed banana.

The practical usefulness of banana quality assessment extends beyond simply identifying ripeness or assigning quality grades. Knowledge of the fruit's maturity stage allows estimation of its remaining shelf life, enabling vendors and consumers to make informed decisions regarding storage, transportation, pricing, and consumption. Bananas approaching full ripeness may be prioritised for immediate sale, while fruits exhibiting excessive defects or advanced ripeness may be discounted, processed into value-added products, or removed from retail shelves to minimise waste. A rule-based recommendation engine therefore provides practical decision support by translating image analysis results into actionable recommendations suitable for commercial and household applications.

Although numerous studies have investigated automated fruit quality assessment, many existing systems primarily focus on a single task, such as ripeness identification or defect detection, without integrating multiple post-harvest assessment functions into a unified platform. Furthermore, many recent approaches rely heavily on deep learning algorithms that require large annotated datasets, extensive computational resources, and specialised hardware for model training and deployment. Such requirements may limit accessibility for small businesses, educational institutions, and users with limited computational resources. There remains a need for an integrated, cost-effective, and transparent image processing system capable of performing ripeness assessment, defect detection, quality grading, shelf-life estimation, recommendation generation, historical data management, and automated reporting using deterministic image processing techniques.

To address these limitations, this project proposes **SmartBanana**, an image processing-based banana ripeness and quality assessment system that integrates twelve functional modules within a unified application. The system acquires banana images through image upload or camera capture before applying image preprocessing techniques to improve image quality. Banana segmentation is subsequently performed using HSV colour space conversion, thresholding, morphological operations, and contour detection. The segmented banana is analysed through colour-based feature extraction to determine ripeness, while defect detection techniques quantify surface imperfections for objective quality grading. Based on the assessment results, SmartBanana estimates the remaining shelf life and generates practical recommendations regarding fruit handling and utilisation. In addition, the system supports batch image processing, video frame analysis, interactive dashboards, PDF report generation, and historical data management to improve inspection efficiency and record keeping. By integrating classical image processing techniques into a comprehensive decision-support platform, SmartBanana aims to provide an accurate, efficient, transparent, and cost-effective solution for automated banana ripeness and quality assessment.

1.2 Objectives/Aims

The objectives of this project are as follows:

1. To develop an image processing-based banana ripeness assessment system capable of automatically analysing banana images and determining the ripeness stage using colour analysis techniques in the HSV colour space.

2. To evaluate banana quality by detecting surface defects, calculating the defect area percentage, and assigning quality grades through rule-based image processing techniques.

3. To generate shelf-life estimation, quality recommendations, analytical visualisations, and automated reports based on the ripeness assessment and quality grading results.

1.3 Motivation

Food waste has become a significant global challenge that affects food security, environmental sustainability, and economic development. According to the Food and Agriculture Organization (FAO), a substantial proportion of food produced for human consumption is lost or wasted throughout the supply chain, with fresh fruits and vegetables representing one of the highest categories of post-harvest losses due to their perishable nature (Food and Agriculture Organization [FAO], 2023). Bananas are particularly susceptible to quality deterioration because their ripening process continues rapidly after harvest. Once bananas become overripe or develop visible defects, their commercial value decreases significantly, often resulting in unnecessary disposal despite the fruit remaining edible. Therefore, improving the efficiency and consistency of banana quality assessment can contribute to reducing food waste while promoting more effective utilisation of agricultural resources.

Current banana quality assessment practices remain largely dependent on manual visual inspection performed by workers, retailers, or consumers. Although experienced inspectors are capable of estimating fruit ripeness and identifying visible defects, the assessment process is inherently subjective and may vary depending on individual judgement, working conditions, lighting environments, and operator experience. In large-scale retail stores, supermarkets, and fruit distribution centres, manual inspection also requires considerable labour and time when processing large quantities of bananas. These limitations may lead to inconsistent grading decisions, inaccurate pricing strategies, delayed inventory management, and increased post-harvest losses. Consequently, there is a growing need for an automated, objective, and cost-effective solution that can assist users in evaluating banana quality consistently using digital image analysis.

The proposed SmartBanana system provides practical value by integrating multiple image processing techniques into a single platform capable of performing banana ripeness assessment, defect detection, quality grading, shelf-life estimation, recommendation generation, and historical data management. Unlike conventional inspection methods, the proposed system applies deterministic image processing algorithms that provide transparent and explainable analysis without requiring large annotated datasets or computationally intensive model training. The integration of batch image processing, dashboard visualisation, automated PDF reporting, and historical record management further enhances operational efficiency by enabling users to analyse multiple bananas, monitor quality trends, and maintain comprehensive inspection records. These features make the system suitable for educational purposes as well as practical deployment in fruit shops, supermarkets, wholesalers, small-scale farms, and research laboratories.

From a commercial perspective, SmartBanana has the potential to support businesses involved in fruit production and distribution by improving inventory management and quality control processes. Objective quality grading enables retailers to categorise bananas according to their market condition, allowing fruits approaching full ripeness to be prioritised for immediate sale while fruits with minor defects may be offered through promotional pricing or alternative processing channels. Shelf-life estimation also assists businesses in planning stock rotation and reducing unnecessary product disposal, thereby minimising financial losses and improving overall operational efficiency. Furthermore, the use of widely available imaging devices and classical image processing techniques reduces implementation costs, making the proposed system accessible to small and medium-sized enterprises that may not possess advanced artificial intelligence infrastructure.

The proposed project also contributes towards the achievement of the **United Nations Sustainable Development Goal (SDG) 12: Responsible Consumption and Production**. SDG 12 promotes the sustainable management and efficient use of natural resources while encouraging the reduction of food loss and waste throughout production and consumption processes (United Nations, 2015). By providing objective ripeness assessment, quality grading, shelf-life estimation, and recommendation functions, SmartBanana assists retailers and consumers in making informed decisions regarding fruit storage, sale, consumption, and processing. These capabilities help maximise the utilisation of edible bananas, reduce unnecessary disposal, and encourage more responsible consumption practices that support sustainable food systems.

In addition to supporting the Sustainable Development Goals, SmartBanana is aligned with the **Malaysia MADANI** framework, particularly the principles of sustainability, innovation, and prosperity. Malaysia MADANI emphasises the application of digital technologies to improve productivity, strengthen economic resilience, and promote sustainable national development (Jabatan Penerangan Malaysia, 2023). The application of image processing techniques for agricultural quality assessment demonstrates how digital innovation can enhance traditional inspection methods while improving operational efficiency and reducing resource wastage. By supporting more accurate fruit quality management and encouraging sustainable consumption practices, SmartBanana contributes towards the broader national vision of developing an innovative, technology-driven, and environmentally responsible society.

Overall, the motivation for developing SmartBanana extends beyond the technical implementation of image processing algorithms. The system addresses practical challenges associated with manual banana quality assessment while providing economic, environmental, and societal benefits. Through objective ripeness assessment, automated defect detection, quality grading, shelf-life estimation, and intelligent recommendation generation, the proposed system has the potential to improve decision-making throughout the banana supply chain. Furthermore, its contribution towards reducing food waste and supporting sustainable agricultural practices demonstrates its relevance to both global sustainability initiatives and Malaysia's long-term digital transformation agenda.



Figure 1.1 SDG9 Industry Innovation and Infrastructure (United Nations, 2015)



Figure 1.2 Malaysia Madani (Jabatan Penerangan Malaysia, 2023)

Literature Review

Scope and research context

Automated fruit inspection sits within well-established computer-vision literature because manual grading is slow, subjective, labour-intensive, and sensitive to operator inconsistency. Recent reviews of conventional imaging for fruit and vegetable quality assessment describe the workflow as a sequence of acquisition, preprocessing, segmentation, feature extraction, and decision-making, and note that RGB cameras remain the dominant sensing modality for many controlled inspection settings because they are inexpensive and sufficiently informative for external attributes such as colour, shape, and visible defects.

For bananas, that focus on visible appearance is especially appropriate. Banana peel colour changes during ripening as chlorophyll degrades, moving from green toward yellow, while advanced ripening is accompanied by surface spotting and darkening. Classical banana studies have therefore treated colour, brown area percentage, and texture as meaningful indicators of maturity and quality rather than as incidental visual cues.

This background strongly supports a non-machine-learning project architecture. In one banana study built around conventional vision features, ripening stages were described using appearance features such as CIELAB colour values, brown spot percentage, and texture. In another, HSV-based banana ripeness classification relied on colour after background removal, showing that peel chromaticity carries substantial discriminative information even before any learning-heavy pipeline is introduced.

2.1 Image segmentation for banana isolation and region extraction

Image segmentation is one of the most fundamental stages in computer vision-based fruit inspection because the accuracy of subsequent analyses, including ripeness assessment, defect detection, and quality grading, depends heavily on the successful isolation of the fruit from its background. In classical image processing, segmentation refers to the partitioning of an image into meaningful regions, enabling the system to distinguish the object of interest from surrounding pixels before quantitative measurements are performed (Baglat et al., 2023). Numerous studies have identified segmentation as the critical first stage in fruit quality inspection, as inaccurate segmentation directly affects the reliability of colour analysis, surface area computation, and defect quantification.

Threshold-based segmentation remains one of the most widely adopted approaches due to its simplicity, computational efficiency, and ease of implementation. Among these techniques, Otsu's thresholding algorithm has become the classical benchmark because it automatically determines the optimal threshold by maximising the variance between foreground and background pixel classes while minimising intra-class variance. This automatic threshold selection eliminates manual parameter tuning and produces highly reproducible segmentation results under controlled imaging conditions. Consequently, Otsu's method has been extensively employed in agricultural image processing, particularly when images are captured under uniform illumination and simple backgrounds. However, previous studies have also reported that the performance of Otsu's algorithm deteriorates when images contain uneven lighting, shadows, reflective surfaces, or complex backgrounds, as these factors distort the grayscale histogram and reduce the separability between object and background regions.

To overcome the limitations associated with grayscale thresholding, colour-space-based segmentation has gained considerable attention in fruit inspection applications. Instead of operating directly on RGB images, many researchers transform images into the HSV (Hue, Saturation, Value) colour space because HSV separates chromatic information from brightness, making colour analysis less sensitive to illumination variations. This characteristic is particularly beneficial for banana ripeness assessment since banana peel colour gradually changes from green to yellow and eventually develops brown or black spots during the ripening process. Previous studies have demonstrated that HSV colour components provide strong discriminative information for identifying ripening stages while simultaneously improving the robustness of fruit segmentation under varying lighting conditions. Consequently, HSV thresholding has become one of the most frequently adopted techniques for classical banana ripeness analysis because it closely resembles human colour perception and simplifies the extraction of colour-based features.

Another widely recognised segmentation technique is watershed segmentation. Unlike thresholding approaches that rely solely on pixel intensity, watershed segmentation interprets an image as a topographical surface, where pixel intensities represent elevation values. The algorithm gradually floods the image from local minima to construct watershed boundaries that separate adjacent regions. This approach is particularly effective when multiple fruits overlap or touch one another, making it suitable for applications involving clustered agricultural products. Nevertheless, watershed segmentation is highly susceptible to over-segmentation because minor image noise often creates numerous local minima, resulting in excessive partitioning of the image. Consequently, most practical implementations employ smoothing filters, marker-controlled watershed algorithms, or morphological preprocessing before applying watershed segmentation to reduce unnecessary region fragmentation.

Clustering-based segmentation techniques have also become increasingly popular for fruit inspection because they are capable of handling images with complex colour distributions. K-means clustering is one of the most frequently adopted unsupervised algorithms, grouping pixels according to colour similarity within perceptually meaningful colour spaces such as CIE L*a*b*. Compared with traditional thresholding, k-means provides greater flexibility when fruit surfaces exhibit non-uniform colours or when backgrounds contain colour variations. Previous research has shown that clustering algorithms are capable of producing more homogeneous segmented regions than simple thresholding; however, they often generate fragmented regions and require additional post-processing techniques, including morphological operations or graph-based region merging, to produce inspection-ready segmentation masks. Consequently, clustering methods generally exhibit greater computational complexity than threshold-based approaches despite offering improved adaptability under challenging imaging conditions.

Based on the existing literature, HSV colour thresholding combined with automatic threshold selection and morphological refinement provides the most appropriate segmentation strategy for the proposed SmartBanana system. Since the project acquires images under relatively controlled environmental conditions using RGB cameras, HSV thresholding offers an excellent balance between segmentation accuracy, computational efficiency, and implementation simplicity. Unlike watershed and clustering techniques, HSV thresholding requires fewer computational resources while maintaining high interpretability, making it particularly suitable for undergraduate image processing projects that emphasise classical computer vision rather than machine learning.

2.2 Contrast Enhancement For Stable Peel And Defect Analysis

Image enhancement constitutes another essential preprocessing stage in classical image processing because the effectiveness of segmentation, feature extraction, and defect analysis depends on the visual quality of the acquired images. Variations in illumination, camera exposure, surface reflections, and shadows frequently reduce image contrast, making it difficult to distinguish fruit boundaries and surface defects accurately. Consequently, numerous studies have highlighted the importance of image enhancement techniques for improving image quality before segmentation and quantitative analysis are performed.

Histogram Equalisation (HE) is one of the earliest and most widely applied global contrast enhancement techniques. The algorithm redistributes pixel intensities across the available dynamic range, thereby improving overall image contrast and making previously indistinguishable features more visible. Due to its computational simplicity and rapid execution, histogram equalisation has been extensively utilised in image processing applications involving medical imaging, industrial inspection, and agricultural quality assessment. Nevertheless, because the transformation is applied globally across the entire image, local contrast differences are often not sufficiently enhanced. Furthermore, excessive histogram stretching may amplify image noise and distort brightness, particularly when images contain large homogeneous regions or significant illumination variation.

Adaptive Histogram Equalisation (AHE) was subsequently introduced to address the shortcomings of global histogram equalisation by performing local contrast enhancement within small neighbourhoods rather than across the entire image. This localised enhancement enables subtle features, including fruit edges, small bruises, and texture variations, to become significantly more distinguishable. Despite these advantages, previous studies have reported that AHE frequently amplifies random image noise within relatively uniform regions because local histograms are independently enhanced throughout the image. In addition, the computational cost of AHE is considerably higher than conventional histogram equalisation, making it less suitable for real-time processing or large image datasets.

To overcome these limitations, Contrast Limited Adaptive Histogram Equalisation (CLAHE) was developed as an improved version of AHE. Instead of allowing unrestricted histogram amplification, CLAHE introduces a clip limit that prevents excessive enhancement within homogeneous image regions. Consequently, CLAHE significantly improves local contrast while suppressing unnecessary noise amplification. Numerous agricultural vision studies have demonstrated that CLAHE effectively enhances subtle fruit surface features, particularly under non-uniform illumination, making it one of the most widely adopted enhancement techniques in modern fruit inspection systems. Although CLAHE requires greater computational resources than conventional histogram equalisation, its superior robustness under challenging lighting conditions generally outweighs the additional processing cost.

Retinex-based enhancement represents another important category of illumination correction techniques. Unlike histogram-based approaches that primarily redistribute pixel intensities, Retinex algorithms estimate the illumination component of an image and separate it from surface reflectance. This process produces images with improved colour constancy and enhanced visibility under uneven lighting conditions. Several studies have demonstrated that Retinex-based algorithms effectively preserve colour fidelity while simultaneously improving local contrast, making them suitable for outdoor agricultural imaging where illumination changes continuously. However, Retinex algorithms

involve substantially greater mathematical complexity and require careful parameter optimisation, making them less practical for undergraduate projects that prioritise simplicity, interpretability, and computational efficiency.

For the proposed SmartBanana system, CLAHE is selected as the primary contrast enhancement technique because it offers the most balanced combination of robustness, image quality improvement, and implementation feasibility. Since banana images may exhibit slight variations in brightness caused by camera positioning or ambient lighting, CLAHE enhances peel colour consistency while preserving subtle bruises and dark spots required for defect detection. Compared with histogram equalisation and Retinex-based methods, CLAHE provides superior local enhancement without introducing excessive computational complexity, making it highly compatible with the subsequent HSV segmentation and morphological processing stages adopted in this project.

2.3 Mathematical Morphology and Connected-component Analysis For Defect Quantification

Mathematical morphology is a fundamental image processing technique that enhances binary images by analysing the geometric structure of objects. Following image segmentation, binary masks frequently contain imperfections such as isolated noise pixels, small holes, irregular boundaries, and fragmented object regions. These artefacts reduce the accuracy of subsequent measurements, including fruit area estimation, contour extraction, colour analysis, and defect quantification. Consequently, mathematical morphology has become a standard post-processing technique in agricultural computer vision to improve segmentation quality before feature extraction and object analysis are performed (Baglat et al., 2023).

Morphological processing is based on the application of predefined structuring elements that probe the geometric properties of binary images. Four fundamental operations form the basis of morphological analysis: erosion, dilation, opening, and closing. Erosion removes boundary pixels, thereby eliminating small isolated objects and reducing object size. Dilation performs the opposite operation by expanding object boundaries and filling narrow gaps between neighbouring pixels. Opening, which consists of erosion followed by dilation, effectively removes small background noise while preserving the overall shape of larger objects. Conversely, closing, which applies dilation followed by erosion, fills small holes and repairs discontinuities within segmented regions. These complementary operations significantly improve the geometric integrity of binary masks, allowing subsequent measurements to represent the true physical characteristics of the fruit surface.

In banana ripeness assessment, morphological operations play a particularly important role because HSV thresholding often produces minor segmentation errors caused by lighting variation, glossy peel reflections, or sensor noise. Small background artefacts may remain connected to the fruit mask, while tiny gaps may appear along fruit boundaries. Morphological opening effectively removes isolated background pixels that are incorrectly classified as fruit, whereas morphological closing reconnects fragmented peel regions and fills internal cavities. The resulting binary mask provides a more accurate representation of the banana, improving contour detection, surface area calculation, and colour percentage analysis. Previous agricultural imaging studies consistently recommend combining

threshold-based segmentation with morphological refinement to increase segmentation robustness and measurement accuracy.

Beyond the four basic operations, more advanced morphological transformations further enhance defect detection. The top-hat transformation extracts bright objects that are smaller than the structuring element, whereas the bottom-hat transformation highlights dark objects against brighter backgrounds. Since overripe bananas commonly develop brown bruises, black spots, and localised dark patches, bottom-hat morphology is particularly valuable because it increases the visibility of these darker surface defects without significantly affecting healthy peel regions. Consequently, defect regions become easier to isolate through thresholding and subsequent quantitative analysis.

Once morphological processing has produced a clean binary mask, Connected Component Analysis (CCA) provides the transition from image segmentation to quantitative feature extraction. Connected Component Analysis assigns a unique label to every connected region within a binary image, allowing individual objects or defect regions to be analysed independently. Rather than treating all detected defects as a single region, CCA separates each bruise or black spot into an individual connected component. This enables the system to determine the number of defects, calculate the area of each defect, measure total damaged surface area, and estimate the percentage of damaged peel relative to the overall banana surface. Such measurements provide objective numerical indicators that directly support quality grading.

Compared with deep learning approaches that automatically infer quality labels from raw images, morphology and connected component analysis provide complete interpretability throughout the inspection pipeline. Every grading decision can be traced to measurable image properties, including segmented fruit area, brown spot percentage, defect count, and colour distribution. This transparency represents a significant advantage for undergraduate image processing projects because each intermediate processing stage can be visualised, validated, and explained scientifically. Furthermore, the methodology closely aligns with the assignment requirement of applying classical image processing techniques, including enhancement, segmentation, morphology, feature extraction, and object analysis, rather than relying on machine learning models.

2.4 Critical Comparison of Existing Approaches

Numerous image processing techniques have been proposed for fruit inspection. However, no single algorithm consistently achieves optimal performance across all imaging environments. The effectiveness of a particular technique depends largely on image acquisition conditions, illumination consistency, background complexity, fruit overlap, and the intended inspection objective. Consequently, previous studies have emphasised selecting algorithms according to the characteristics of the inspection environment rather than adopting the most sophisticated method available.

Threshold-based segmentation methods remain among the most popular approaches because they are computationally efficient, straightforward to implement, and highly interpretable. Otsu's thresholding automatically determines an optimal segmentation threshold and performs exceptionally well when images are captured against uniform backgrounds under stable lighting conditions. Nevertheless, the algorithm becomes increasingly unreliable when images exhibit non-uniform illumination, shadows, reflections, or overlapping intensity distributions. Under such circumstances, segmentation errors may significantly reduce the accuracy of subsequent colour analysis and defect measurement.

Consequently, Otsu's algorithm is generally recommended only after suitable colour-space conversion or image enhancement has been applied.

Watershed segmentation offers substantially greater flexibility by modelling images as topographical surfaces and identifying boundaries through simulated flooding processes. This characteristic enables watershed algorithms to separate touching or overlapping fruits that cannot easily be isolated using conventional thresholding. Despite these advantages, watershed segmentation frequently suffers from over-segmentation because even minor image noise generates additional local minima, producing an excessive number of segmented regions. Practical implementations therefore require additional preprocessing, including Gaussian smoothing, marker generation, or morphological filtering, increasing computational complexity and implementation difficulty.

Clustering algorithms, particularly k-means clustering, represent another widely investigated segmentation strategy. Instead of relying on manually selected thresholds, k-means groups pixels according to colour similarity, allowing the algorithm to accommodate more complex colour distributions and varying backgrounds. Studies have demonstrated that clustering techniques generally outperform fixed thresholding when fruit surfaces exhibit heterogeneous colour characteristics. However, clustering algorithms often produce fragmented regions, ignore spatial continuity between neighbouring pixels, and require subsequent region merging or morphological refinement before meaningful quantitative measurements can be performed. As a result, clustering approaches typically require greater computational resources than threshold-based segmentation while providing only moderate improvements under controlled imaging conditions.

Image enhancement techniques exhibit similar trade-offs. Global Histogram Equalisation provides rapid improvement of overall image contrast but frequently fails to enhance local details associated with bruises, texture variations, and subtle colour transitions. Adaptive Histogram Equalisation improves local contrast considerably but often amplifies image noise within homogeneous regions. CLAHE successfully addresses these shortcomings by limiting excessive contrast amplification while preserving local image detail. Consequently, CLAHE has become the preferred enhancement technique in numerous agricultural vision applications involving variable illumination and subtle surface defects.

The literature therefore indicates that algorithm selection should prioritise simplicity, robustness, and interpretability over algorithmic complexity whenever images are acquired under controlled laboratory conditions. Sophisticated segmentation algorithms provide measurable advantages primarily when imaging environments involve overlapping fruits, highly variable illumination, or complex backgrounds. In contrast, classical pipelines combining CLAHE, HSV thresholding, morphological refinement, contour analysis, and connected component analysis consistently provide reliable performance for single-fruit inspection while remaining computationally efficient and easy to interpret. These characteristics closely match the objectives of the proposed SmartBanana system and the emphasis of the assignment on classical image processing methodologies.

Table 2.1 Comparison of Image Segmentation Techniques

Algorithm / System	**Example of Previous Work *(insert as citation)***	**Advantages**	**Limitations**	**Remarks**
Algorithm 1: Otsu Thresholding	Otsu (1979) proposed an automatic histogram-based threshold selection method that remains one of the most widely adopted segmentation techniques in image processing (Otsu, 1979).	Fast, automatic, computationally efficient, reproducible, simple implementation.	Sensitive to uneven illumination and non-bimodal histograms.	Highly suitable because banana images are captured under controlled conditions.
Algorithm 2: HSV Thresholding	Saroinsong et al. (2020) proposed banana ripeness classification using the HSV colour space and a Nearest Centroid Classifier, demonstrating that HSV effectively distinguishes banana ripeness based on peel colour changes. Baglat et al. (2023) also identified HSV colour analysis as one of the	Robust against illumination variation, directly analyses banana peel colour.	Threshold values may require calibration for different environments.	Most suitable because ripeness depends primarily on peel colour changes.

	most widely adopted classical approaches for banana ripeness assessment.			
Algorithm 3: Watershed Segmentation	Vincent and Soille (1991) introduced watershed segmentation for separating connected regions based on topographical image analysis (Vincent & Soille, 1991).	Excellent for separating touching objects.	High risk of over-segmentation; requires extensive preprocessing.	Suitable only when multiple bananas overlap.
Algorithm 4: K-Means Clustering	Pham and Lee (2015) applied K-Means clustering in the Lab* colour space for fruit defect segmentation before graph-based region merging (Pham & Lee, 2015).	Handles complex colour distributions and heterogeneous surfaces.	Computationally intensive; frequently requires post-processing.	Suitable for uncontrolled environments but unnecessary for single-fruit analysis.

Table 2.2 Comparison of Contrast Enhancement Techniques

Algorithm / System	**Example of Previous Work**	**Advantages**	**Limitations**	**Remarks**

	(insert as citation)			
Algorithm 1: Histogram Equalisation (HE)	Histogram Equalisation has long been regarded as the standard global contrast enhancement technique for image processing applications (Pizer et al., 1987).	Fast, simple implementation , improves global image contrast.	Poor local enhancement; may amplify brightness inconsistently.	Suitable only under uniform lighting.
Algorithm 2: Adaptive Histogram Equalisation (AHE)	Pizer et al. (1987) introduced Adaptive Histogram Equalisation to improve local image contrast through neighbourhood-based enhancement (Pizer et al., 1987).	Enhances local image details effectively.	Amplifies image noise; higher computational complexity.	Useful but not recommended for final implementation .
Algorithm 3: CLAHE	Zuiderveld (1994) proposed CLAHE as an improved version of AHE by introducing a contrast clipping mechanism to	Enhances local contrast while limiting noise amplification; robust under varying illumination.	Requires parameter tuning; slightly higher computational cost.	Selected for the proposed system because it provides the best balance between robustness and

	prevent excessive enhancement (Zuiderveld, 1994).			computational efficiency.
Algorithm 4: Retinex Enhancement	Retinex theory has been widely investigated for illumination correction by separating image reflectance from illumination, producing enhanced colour constancy under complex lighting conditions (Jobson et al., 1997; Rahman et al., 2004).	Excellent illumination correction and colour constancy.	Computationally complex and difficult to optimise.	Better suited to advanced research rather than undergraduate coursework.

Table 2.3 Comparison of Defect Detection Algorithms

Algorithm / System	**Example of Previous Work *(insert as citation)***	**Advantages**	**Limitations**	**Remarks**
Algorithm 1: Threshold-Based Defect Detection	Classical fruit inspection systems commonly use grayscale or colour thresholding to isolate bruises	Simple implementation, fast computation, suitable for	Sensitive to illumination changes, may miss subtle bruises.	Suitable for detecting dark brown and black banana spots after

	and dark spots (Baglat et al., 2023).	obvious defects.		HSV segmentation.
Algorithm 2: Morphological Processing	Serra (1982) introduced mathematical morphology for analysing object shapes using erosion, dilation, opening and closing operations (Serra, 1982).	Removes segmentation noise, repairs fragmented objects, and improves defect localisation.	Performance depends on structuring element selection, may remove very small defects if parameters are unsuitable.	Essential post-processing stage after thresholding to improve defect mask quality.
Algorithm 3: Connected Component Analysis (CCA)	He et al. (2017) reviewed Connected Component Analysis as a standard approach for identifying individual regions within binary images (He et al., 2017).	Counts individual defects, measures defect area accurately, supports quantitative quality grading.	Requires accurate binary segmentation beforehand, cannot distinguish defect types without additional analysis.	Selected because SmartBanana grades bananas according to total damaged surface percentage rather than merely detecting defects.

2.5 Implications for the Proposed SmartBanana System

The reviewed literature demonstrates that classical image processing techniques remain highly effective for fruit ripeness assessment when images are captured under controlled acquisition conditions. Although recent studies increasingly employ deep learning models such as Convolutional Neural Networks (CNNs), YOLO variants, and Transformer-based architectures to achieve state-of-the-art classification performance, these approaches generally require large annotated

datasets, extensive computational resources, and specialised hardware for model training and inference. In contrast, conventional computer vision techniques continue to offer significant advantages in terms of computational efficiency, transparency, ease of implementation, and interpretability, particularly for applications where external fruit characteristics are the primary indicators of quality.

For bananas, peel colour transition represents one of the most reliable visual indicators of physiological ripening. As chlorophyll gradually degrades during maturation, the peel changes progressively from green to yellow before developing brown and black spots associated with over-ripening and tissue degradation. Numerous studies have therefore utilised colour information as the principal feature for ripeness determination, while surface blemishes, bruises, and dark spots provide complementary indicators of fruit quality. Since these characteristics are externally visible, they can be measured accurately using RGB images without requiring expensive hyperspectral or multispectral imaging equipment. Consequently, conventional RGB-based image processing remains an appropriate and academically justified solution for undergraduate fruit inspection systems.

Based on the findings presented throughout this literature review, the proposed SmartBanana system adopts a complete end-to-end classical image processing pipeline consisting of image acquisition, image preprocessing, fruit segmentation, ripeness analysis, defect detection, quality grading, and result visualisation. Initially, captured images undergo resizing and noise reduction using Gaussian filtering, followed by CLAHE to improve local contrast while preserving subtle peel texture and surface defects. The enhanced image is subsequently converted into the HSV colour space, where colour thresholding isolates the banana from the background more effectively than conventional RGB thresholding. Morphological opening and closing operations refine the segmentation mask by removing isolated noise and repairing fragmented fruit boundaries before contour detection identifies the primary fruit region. Finally, connected component analysis quantifies individual defect regions, allowing defect area percentage and surface damage measurements to be incorporated into a rule-based quality grading algorithm.

This processing pipeline directly satisfies the core functional requirements outlined in the assignment specification, including image enhancement, filtering, segmentation, feature extraction, morphology, and object analysis. Furthermore, the proposed implementation extends beyond the minimum requirements by incorporating additional functionalities such as dashboard visualisation, bulk image processing, PDF report generation, shelf-life estimation, and recommendation support. These supplementary modules transform the prototype into a practical fruit quality assessment system suitable for commercial and educational applications while simultaneously demonstrating additional technical effort expected for excellent coursework performance.

An additional advantage of the proposed methodology is its complete interpretability. Unlike deep learning approaches, which frequently operate as "black-box" models, every processing stage within the SmartBanana pipeline produces measurable intermediate outputs that can be visualised and verified. Examples include enhanced images, binary segmentation masks, fruit contours, colour percentage distributions, detected defect regions, connected component labels, and calculated damage percentages. This transparency enables users to understand precisely how the final ripeness classification and quality grade are generated, thereby increasing confidence in the inspection results and facilitating debugging, optimisation, and future system enhancement. Such explainability is

particularly valuable in educational settings where understanding the underlying algorithms is equally important as achieving high prediction accuracy.

Overall, the reviewed literature provides strong theoretical support for the proposed SmartBanana system. By integrating CLAHE, HSV colour analysis, threshold-based segmentation, mathematical morphology, contour extraction, connected component analysis, and rule-based grading within a unified inspection pipeline, the proposed system achieves an effective balance between computational efficiency, measurement accuracy, algorithm transparency, and implementation feasibility. These characteristics closely align with the objectives of the Image Processing module and provide a scientifically justified foundation for the methodology presented in the subsequent chapter.

2.6 Research Gap

Despite considerable progress in automated fruit inspection, several limitations remain evident within the existing literature. First, the majority of recent studies concentrate primarily on ripeness classification while paying relatively little attention to simultaneous defect detection and overall quality grading. Although colour-based maturity assessment has been extensively investigated, quantitative measurement of bruises, black spots, and surface damage is often omitted or treated as a secondary objective. Consequently, many proposed systems are capable of determining ripeness stages but cannot comprehensively evaluate the commercial quality of individual fruits.

Second, most published systems employ deep learning techniques that require extensive annotated datasets and computationally intensive model training. While these methods achieve remarkable classification accuracy, they are generally unsuitable for educational environments that emphasise understanding of classical image processing algorithms. Furthermore, the computational requirements of deep learning models may limit their deployment on low-cost hardware and reduce system interpretability. In contrast, relatively few studies investigate fully classical computer vision pipelines capable of producing accurate, explainable, and computationally efficient fruit inspection systems.

Third, many existing studies perform evaluation using only individual images collected under laboratory conditions. Features such as batch image analysis, automated reporting, dashboard visualisation, shelf-life estimation, and recommendation support remain uncommon despite their practical importance for agricultural businesses and fruit retailers. Consequently, many published solutions remain research prototypes rather than complete end-user applications capable of supporting operational decision-making.

Finally, explainability remains a significant challenge in modern computer vision systems. Deep learning models frequently provide only final classification outputs without revealing the underlying visual evidence supporting each decision. Such limited interpretability may reduce user confidence and complicate model validation, particularly in educational contexts where algorithm understanding forms a major learning objective. Classical image processing techniques address this limitation by allowing every intermediate processing stage to be visualised and quantitatively analysed.

Therefore, the proposed SmartBanana system addresses these research gaps by integrating ripeness assessment, defect quantification, quality grading, shelf-life prediction, recommendation support, and comprehensive reporting within a single classical image processing framework. Unlike many previous studies, every decision generated by the proposed system can be directly explained through

measurable image features, making the system both scientifically interpretable and practically applicable.

Methodology

3.1 System flowchart

The proposed SmartBanana system follows a structured image processing pipeline consisting of image acquisition, preprocessing, banana segmentation, ripeness analysis, defect detection, quality grading, shelf-life prediction, recommendation generation, and report management. Rather than relying on deep learning models, the proposed methodology employs deterministic image processing algorithms that provide transparent and explainable decision-making while maintaining low computational complexity. Each processing stage generates intermediate outputs that can be visualised and verified, allowing users to understand how the final quality assessment is produced. The overall workflow is illustrated in Figure 3.1.

https://drive.google.com/file/d/1WV2_a86RMfLL39SvkFTrzhnMuGygQYTl/view?usp=sharing

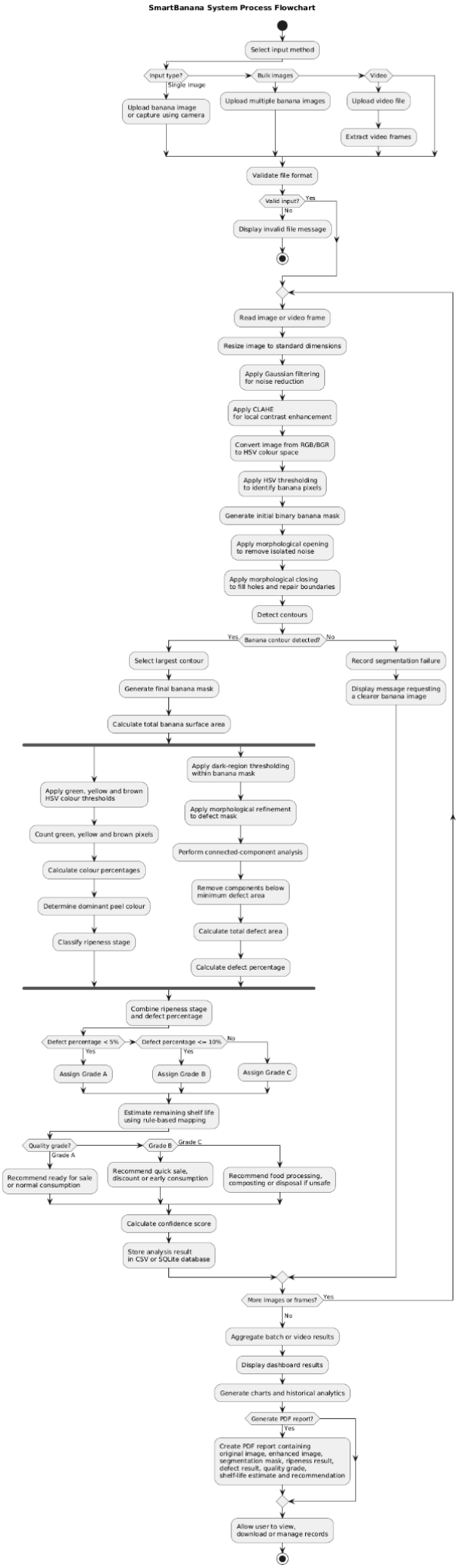

SmartBanana System Process Flowchart
Select input method
Input type?
Single image
Bulk images
Video
Upload banana image or capture using camera
Upload multiple banana images
Upload video file
Extract video frames
Validate file format
Valid input?
Yes
No
Display invalid file message
Read image or video frame
Resize image to standard dimensions
Apply Gaussian filtering for noise reduction
Apply CLAHE for local contrast enhancement
Convert image from RGB/BGR to HSV colour space
Apply HSV thresholding to identify banana pixels
Generate initial binary banana mask
Apply morphological opening to remove isolated noise
Apply morphological closing to fill holes and repair boundaries
Detect contours
Banana contour detected?
Yes
No
Select largest contour
Generate final banana mask
Calculate total banana surface area
Record segmentation failure
Display message requesting a clearer banana image
Apply green, yellow and brown HSV colour thresholds
Count green, yellow and brown pixels
Calculate colour percentages
Determine dominant peel colour
Classify ripeness stage
Apply dark-region thresholding within banana mask
Apply morphological refinement to defect mask
Perform connected-component analysis
Remove components below minimum defect area
Calculate total defect area
Calculate defect percentage
Combine ripeness stage and defect percentage
Defect percentage < 5%
Defect percentage <= 10%
No
Yes
Yes
Assign Grade A
Assign Grade B
Assign Grade C
Estimate remaining shelf life using rule-based mapping
Quality grade?
Grade A
Grade B
Grade C
Recommend ready for sale or normal consumption
Recommend quick sale, discount or early consumption
Recommend food processing, composting or disposal if unsafe
Calculate confidence score
Store analysis result in CSV or SQLite database
More images or frames?
Yes
No
Aggregate batch or video results
Display dashboard results
Generate charts and historical analytics
Generate PDF report?
Yes
Create PDF report containing original image, enhanced image, segmentation mask, ripeness result, defect result, quality grade, shelf-life estimate and recommendation
Allow user to view, download or manage records

Figure 3.1: SmartBanana System Architecture and Image Processing Flowchart.

Figure 3.1 illustrates the complete processing pipeline implemented in the SmartBanana system. The workflow begins when the user uploads an image from the local device or captures a new banana image using the system camera. The acquired image is then forwarded to the image preprocessing stage, where resizing, Gaussian filtering, and Contrast Limited Adaptive Histogram Equalisation (CLAHE) are applied. These operations improve image quality by standardising image dimensions, suppressing sensor noise, and enhancing local contrast, thereby producing a more reliable input for subsequent image analysis.

Following preprocessing, the enhanced RGB image is converted into the HSV colour space. Unlike RGB, HSV separates colour information from brightness, making the segmentation process less sensitive to illumination changes. HSV thresholding is then applied to distinguish banana pixels from the background based on predefined colour ranges corresponding to banana peel colours.

The resulting binary mask may still contain isolated noise, small holes, and fragmented object boundaries. Therefore, morphological opening and closing operations are applied to remove small artefacts while repairing discontinuities within the segmented banana. After morphological refinement, contour detection is performed to identify all connected objects in the binary image. Since only one banana is analysed at a time, the largest contour is selected as the banana object while smaller background objects are discarded. This process produces an accurate banana mask that serves as the basis for all subsequent analyses.

Once the banana has been successfully segmented, the system simultaneously performs ripeness analysis and defect detection. During ripeness analysis, the segmented banana is analysed using HSV colour thresholds corresponding to green, yellow, and brown peel colours. The percentage of pixels belonging to each colour category is calculated to objectively determine the maturity stage of the banana. Based on predefined decision rules, the system classifies the banana into one of several ripeness categories, including unripe, partially ripe, ripe, or overripe.

At the same time, defect detection is performed by identifying dark regions that correspond to bruises, black spots, fungal infection, or surface deterioration. Binary thresholding is applied to isolate potential defects, followed by morphological refinement to eliminate false detections. Connected Component Analysis is subsequently employed to identify each defect as an independent connected region. The total defect area is calculated as a percentage of the entire banana surface, providing an objective numerical indicator of fruit quality.

The ripeness classification and defect percentage are subsequently integrated within a rule-based quality grading module. Rather than relying on subjective human judgement, the grading rules combine colour information and surface damage measurements to classify bananas into predefined quality categories. This transparent grading mechanism ensures that every decision can be directly explained through measurable image features.

Following quality grading, the system estimates the remaining shelf life according to the detected ripeness stage and quality condition. A recommendation engine then generates practical suggestions for storage, consumption, selling, or processing based on the predicted shelf life. For example,

bananas that are fully ripe may be recommended for immediate consumption, whereas overripe bananas may be suggested for baking or smoothie preparation.

Finally, all inspection results are presented through the dashboard module, where users can review the original image, segmentation results, ripeness classification, defect percentage, quality grade, shelf-life estimation, and recommendations. The system also supports batch image processing, video analysis, PDF report generation, and historical record management, enabling users to analyse multiple banana images efficiently while maintaining complete inspection records for future reference.

Overall, the proposed methodology integrates image enhancement, segmentation, feature extraction, morphology, object analysis, and decision-making into a unified image processing pipeline. Compared with manual inspection, the proposed workflow provides more objective, consistent, and explainable quality assessment while satisfying the assignment requirement of applying classical image processing techniques throughout the system implementation.

3.2 Description of dataset

The SmartBanana system uses a combination of publicly available banana images and self-collected images to support algorithm calibration, functional testing, and experimental evaluation. Although the proposed system is based primarily on classical image processing rather than supervised machine learning, a varied image collection remains necessary to determine appropriate HSV threshold ranges, evaluate segmentation accuracy, test defect detection, and assess the reliability of the final ripeness and quality grading rules.

The selected public image source is the **FreshCheck Fruit Ripeness Classification Dataset**, which contains banana and tomato images organised according to different ripeness and quality conditions. Only the banana subset is used in the SmartBanana project because the system is specifically designed for banana ripeness and surface quality assessment. The dataset provides images representing different peel colour conditions, including green, partially yellow, fully yellow, and heavily spotted or overripe bananas. These visual differences are suitable for evaluating the proposed colour-based ripeness analysis process.

Dataset source: FreshCheck Fruit Ripeness Classification Dataset
Dataset platform: Kaggle
Dataset URL:
https://www.kaggle.com/datasets/orofinadedamola/freshcheck-fruit-ripeness-classification-dataset
Selected subset: Banana images only
Image formats: JPG, JPEG, and PNG
Banana images: Approximately 13,478 images
Ripeness classes: Four classes
Planned data division: 80% testing and calibration, 10% validation, and 10% final evaluation

The dataset is not used to train an artificial intelligence model. Instead, representative images from each ripeness class are examined to determine suitable colour thresholds and to test whether the classical image processing pipeline can distinguish the major banana ripeness stages. This approach remains consistent with the assignment requirement to apply image enhancement, filtering, segmentation, feature extraction, morphology, and object analysis techniques.

3.2.1 Dataset Categories

The banana images are grouped into four general ripeness categories according to their visible peel characteristics. The categories used by SmartBanana are shown in Table 3.1.

Category	Visible Peel Characteristics	SmartBanana Output
Unripe	Peel is predominantly green with little or no yellow colour	Unripe
Ripe	Peel is predominantly yellow with little or no brown spotting	Ripe
Overripe	Peel is yellow with noticeable brown spots indicating advanced ripening	Overripe
Spoiled	Peel contains extensive black or dark brown regions, decay, or severe deterioration	Spoiled

Table 3.1: Banana Ripeness Categories

The banana images are organised into four categories according to the FreshCheck dataset: unripe, ripe, overripe, and spoiled. These categories represent different stages of banana maturity and deterioration based on external peel appearance. The SmartBanana system uses these labelled images to evaluate the effectiveness of the proposed image processing algorithms. During processing, the colour distribution of the banana peel and the percentage of surface defects are analysed to determine which category most closely matches the input image. The spoiled category differs from the overripe category because it represents bananas with severe surface deterioration, extensive blackening, fungal growth, or decay that significantly reduces their quality and remaining shelf life.

3.2.2 Self-Collected Images

In addition to the public dataset, self-collected banana images are included to evaluate the prototype under realistic conditions. These images are captured using ordinary mobile phone cameras and computer webcams. Bananas at different ripeness stages are photographed under several lighting conditions, image distances, camera angles, and backgrounds.

The self-collected images are used for the following purposes:

1. Testing whether the system can process images captured outside the original dataset environment.
2. Evaluating the effect of illumination variation on HSV thresholding.
3. Testing the effectiveness of Gaussian filtering and CLAHE enhancement.
4. Assessing whether bruises, brown spots, and black surface regions can be detected.
5. Evaluating single-image, batch-image, and video-frame processing.

6. Identifying cases in which the segmentation or grading rules require further parameter adjustment.

To reduce unnecessary variability, the recommended image acquisition environment uses a plain contrasting background, sufficient lighting, minimal shadow, and a single banana positioned near the centre of the image. The entire banana should be visible and should not be heavily covered by other objects. Images that are excessively blurred, underexposed, overexposed, or incomplete may be rejected by the input validation module.

3.2.3 Image Acquisition Requirements

For reliable processing, each image should satisfy the following acquisition requirements:

- The input must use the JPG, JPEG, or PNG format.
- At least one clearly visible banana must appear in the image.
- The fruit should occupy a substantial portion of the image.
- The entire banana surface should be visible whenever possible.
- The image should contain sufficient lighting to distinguish green, yellow, and brown peel regions.
- Severe motion blur should be avoided.
- A background that contrasts with the banana peel is recommended.
- The image should not contain strong yellow objects that may be incorrectly included within the segmentation mask.

All accepted images are converted into a standard internal format before processing. Colour images are read as three-channel arrays, resized to a fixed maximum dimension, and stored temporarily as NumPy arrays for OpenCV processing. Resizing standardises the computational workload while preserving the original aspect ratio.

3.2.4 Data Structure

Each image analysis produces a structured record containing the input details, extracted image features, and final assessment results. The proposed data structure is shown in Table 3.2.

Field Name	Data Type	Description
record_id	Integer/String	Unique identifier assigned to each analysis record
file_name	String	Original name of the uploaded image or video frame
input_type	String	Indicates whether the input is an image, batch image, camera image, or video frame
date_time	DateTime	Date and time when the analysis was performed

image_width	Integer	Width of the processed image in pixels
image_height	Integer	Height of the processed image in pixels
fruit_area	Integer	Number of pixels contained within the final banana mask
green_pixels	Integer	Number of segmented banana pixels classified as green
yellow_pixels	Integer	Number of segmented banana pixels classified as yellow
brown_pixels	Integer	Number of segmented banana pixels classified as brown
green_percentage	Float	Percentage of banana surface classified as green
yellow_percentage	Float	Percentage of banana surface classified as yellow
brown_percentage	Float	Percentage of banana surface classified as brown
ripeness_stage	String	Final ripeness classification: unripe, ripe, overripe or spoiled.
defect_count	Integer	Number of valid connected defect regions
defect_area	Integer	Total area of all detected surface defects in pixels
defect_percentage	Float	Percentage of the banana surface classified as defective
quality_grade	String	Final quality grade assigned by the rule-based grading module
shelf_life_days	Integer	Estimated number of remaining shelf-life days
recommendation	String	Suggested action, such as sale, quick consumption, discounting, or processing
processing_time	Float	Total execution time required to analyse the image

status	String	Indicates whether processing was successful or whether an error occurred

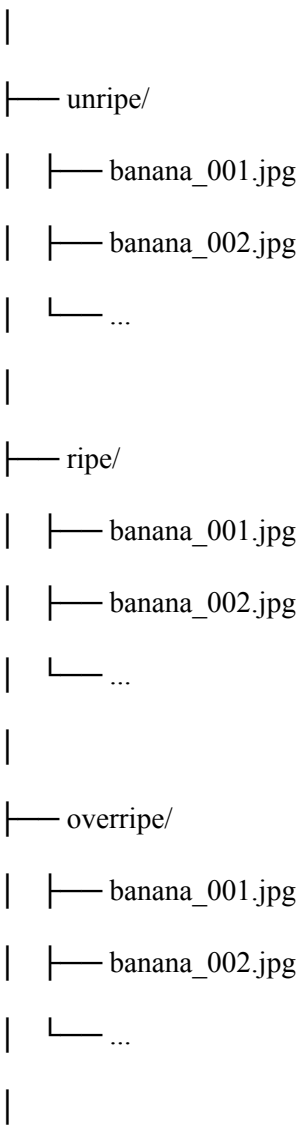
Table 3.2: SmartBanana Data Dictionary

The data dictionary ensures that every inspection record follows a consistent structure. It also supports dashboard visualisation, historical comparison, batch processing summaries, and automated PDF report generation. The current project scope allows records to be stored in CSV format or in a lightweight SQLite database. The existing report identifies data management, dashboard analysis, batch processing, video processing, and PDF reporting as integrated parts of the proposed system.

3.2.5 Directory Organisation

The image collection should be organised into separate folders according to ripeness category. A proposed directory structure is shown below.

dataset/



```
├── spoiled/
│   ├── banana_001.jpg
│   ├── banana_002.jpg
│   └── ...
│
└── self_collected/
    ├── controlled_background/
    ├── variable_lighting/
    ├── bruised_samples/
    └── video_frames/
```

This structure allows images to be selected according to their expected ripeness condition during system calibration and evaluation. It also simplifies the calculation of confusion matrices and classification accuracy during the result and discussion stage.

3.2.6 Data Preparation

Before an image is passed to the main processing pipeline, several preparation operations are performed. First, the file extension is checked to ensure that the uploaded file is supported. The image is then decoded and verified to confirm that it is not corrupted. Images with invalid dimensions or missing colour channels are rejected.

Accepted images are resized while preserving the aspect ratio. A maximum dimension such as 800 or 1,024 pixels may be used to reduce processing time without removing essential peel details. The resized image is stored in BGR format because this is the default representation used by OpenCV. It is subsequently converted into other colour spaces depending on the processing stage.

A Gaussian filter is applied to reduce minor image noise. CLAHE is then applied to the luminance component of the image to improve local contrast without excessively altering peel colour information. After enhancement, the image is converted into HSV format for banana segmentation, ripeness analysis, and defect detection.

No artificial alteration of the final test images is performed. Operations such as rotation, brightness modification, or colour adjustment may be used during early parameter calibration, but the final evaluation should contain unmodified images to ensure that the reported performance reflects realistic system behaviour.

3.2.7 Ground-Truth Labelling

To evaluate the accuracy of the SmartBanana system, each selected test image should be assigned a ground-truth ripeness label. The ground-truth category may be determined through agreement

between group members using the visible peel characteristics shown in Table 3.1. Images with uncertain or ambiguous ripeness conditions should be reviewed collectively before a final label is assigned.

Defect evaluation requires a separate ground-truth procedure. For a representative subset of images, visible defect regions may be manually outlined using an image annotation tool. The manually marked regions can then be compared with the automatically generated defect mask. This comparison enables the calculation of segmentation metrics such as Intersection over Union, precision, recall, and Dice coefficient.

The use of manually reviewed labels provides a reference against which the automated output can be evaluated. However, the labels remain based on visible external characteristics and should not be interpreted as laboratory measurements of internal fruit condition.

3.2.8 Sample Dataset Image

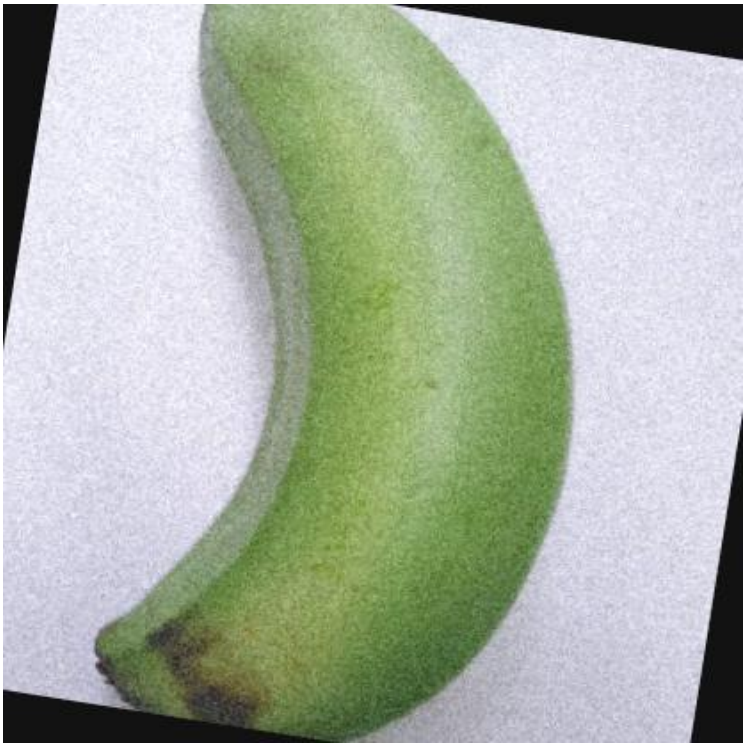


Figure 3.2(a): Unripe banana sample



Figure 3.2(b): Ripe banana sample



Figure 3.2(c): Overripe banana sample



Figure 3.2(d): Spoiled banana sample

Figure 3.2: Sample banana images representing the four ripeness categories used for SmartBanana evaluation.

3.2.9 Dataset Limitations

The selected dataset and self-collected image collection contain several limitations. The images may differ in camera quality, resolution, background, illumination, banana variety, and viewing angle. Furthermore, the provided class labels may not follow an identical physiological ripeness scale across all image sources. External appearance may also be affected by bruising, shadows, surface reflections, and dirt, which can resemble natural brown ripening spots.

The system is therefore designed primarily for clearly visible single bananas captured under reasonably controlled conditions. Results obtained from heavily overlapping bananas, incomplete fruit images, extreme lighting, or complex backgrounds may be less reliable. These limitations are addressed through input guidance, contrast enhancement, morphological refinement, minimum-area filtering, and clear reporting of segmentation failures.

Overall, the combined public and self-collected image collection provides sufficient visual variation to calibrate and evaluate the SmartBanana image processing pipeline. The dataset supports the assessment of ripeness classification, banana segmentation, surface defect quantification, quality grading, processing efficiency, and robustness under different acquisition conditions. This methodology is aligned with the report's proposed classical processing approach and its focus on transparent, measurable image features.

3.3 Applications of the algorithm(s)

The SmartBanana system integrates a series of classical image processing algorithms to perform banana ripeness classification and quality assessment. Each algorithm is responsible for a specific stage of the processing pipeline, beginning with image acquisition and ending with quality recommendation and report generation. Unlike machine learning approaches, the proposed methodology uses deterministic algorithms that allow every intermediate processing result to be visualised and interpreted. Consequently, the final ripeness classification and quality grading can be directly explained using measurable image features such as colour distribution, defect area, and object geometry.

3.3.1 Image Acquisition

The first stage of the SmartBanana system acquires banana images from the user. Three acquisition methods are supported:

- Single image upload
- Batch image upload
- Video file processing

The system first validates the uploaded file format before decoding the image using OpenCV.

Supported formats include:

- JPG
- JPEG
- PNG
- MP4 (video)

Invalid or corrupted files are rejected before further processing.

The decoded image is stored as a three-channel BGR image because OpenCV internally represents colour images using the BGR colour space.

Sample Input: banana001.jpg
Sample Output:

- Loaded image
- Width = 1024 pixels
- Height = 768 pixels
- Channels = 3

Pseudocode:

Algorithm 1 Image Acquisition Input: Image File

Output: BGR Image

Begin

Read image

If image cannot be loaded

Display error

Terminate

Return image

End

3.3.2 Image Preprocessing

Before banana segmentation, the acquired image undergoes preprocessing to improve image quality and reduce unwanted variations.

Three preprocessing operations are performed:

1. Image resizing
2. Gaussian filtering
3. CLAHE enhancement

Image resizing standardises image dimensions while reducing computational cost.

Gaussian filtering removes small sensor noise without significantly affecting object boundaries.

The Gaussian filter is defined as

$$G(x,y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$

where

- x, y represent pixel coordinates
- σ controls smoothing strength

A 5 × 5 Gaussian kernel is applied.

After noise removal, Contrast Limited Adaptive Histogram Equalisation (CLAHE) enhances local contrast while preventing excessive amplification of image noise.

Unlike standard Histogram Equalisation, CLAHE divides the image into small tiles before performing local histogram equalisation.

This improves visibility of

- bruises
- peel texture
- dark spots
- colour transitions

without significantly altering banana colour.

Sample Input: Original RGB image
Sample Output: Enhanced image with improved contrast

Pseudocode:

Algorithm 2 Image Preprocessing

Input: Original Image

Output: Enhanced Image

Resize image

Apply Gaussian Blur

Apply CLAHE

Return Enhanced Image

3.3.3 HSV Colour Conversion

Since RGB is highly affected by illumination, the enhanced image is converted into HSV colour space.

HSV separates

- Hue
- Saturation

- Value

allowing colour information to be analysed independently from brightness.

Hue values represent the actual peel colour.

Green bananas exhibit lower hue values.

Yellow bananas exhibit larger hue values.

Brown and black regions produce low saturation and low brightness.

OpenCV performs colour conversion using

```
cv2.cvtColor(image, cv2.COLOR_BGR2HSV)
```

Pseudocode:

Algorithm 3 HSV Conversion Input: Enhanced Image   Convert RGB to HSV Return HSV Image

3.3.4 Banana Segmentation Using HSV Thresholding

Banana segmentation isolates the fruit from the surrounding background.

The HSV image is thresholded according to predefined lower and upper colour limits.

Each pixel satisfies

$$Lower \leq Pixel \leq Upper$$

Pixels inside the threshold become white

Pixels outside become black

The resulting binary image contains the banana mask.

HSV thresholding performs better than RGB thresholding because brightness variation has less influence on Hue.

Sample Output: Binary banana mask

Pseudocode:

Algorithm 4 HSV Thresholding

Input: HSV Image

For every pixel

    If HSV within threshold

        Mask = White

    Else

        Mask = Black

Return Binary Mask

3.3.5 Morphological Processing

Thresholding often introduces

- isolated pixels
- holes
- fragmented boundaries

Morphological Opening removes isolated foreground noise.

Opening consists of

$$Opening = Erosion \oplus Dilation$$

Morphological Closing repairs small holes.

$$Closing = Dilation \ominus Erosion$$

The structuring element is a 5 × 5 elliptical kernel.

These operations produce a cleaner banana mask before contour extraction.

Sample Output: Clean segmentation mask

Pseudocode:

```
Algorithm 5 Morphology

Input: Binary Mask


Apply Opening

Apply Closing

Return Clean Mask
```

3.3.6 Contour Detection and Banana Isolation

Contours are extracted from the refined binary mask.

Each contour represents one connected object.

Since only one banana is analysed at a time, the largest contour is selected.

The contour is used to calculate

- banana area
- banana boundary
- banana mask

Pixels outside the contour are discarded.

This ensures that background pixels do not influence ripeness analysis.

Pseudocode:

```
Algorithm 6 Contour Detection

Input: Clean Binary Mask


```

Detect Contours

Find Largest Contour

Generate Banana Mask

Return Banana

3.3.7 Ripeness Analysis

The segmented banana is analysed according to peel colour.

Three HSV masks are generated:

- Green
- Yellow
- Brown

The percentage of each colour is calculated by

$$Percentage = \frac{ColourPixels}{TotalBananaPixels} \times 100$$

The dominant colour determines the ripeness stage.

Example:

Green	Yellow	Brown	Result
80%	18%	2%	Unripe
15%	82%	3%	Ripe
5%	55%	40%	Overripe
2%	15%	83%	Spoiled

Pseudocode:

Algorithm 7 Ripeness Analysis

Input: Banana Mask

Calculate Green %

Calculate Yellow %

Calculate Brown %

Determine Dominant Colour

Return Ripeness Stage

3.3.8 Defect Detection

Dark regions are extracted from the segmented banana.

Binary thresholding identifies potential defects.

Connected Component Analysis labels every connected defect.

Small components below a minimum area threshold are removed.

The total defect area is

$$Defect\ Percentage = \frac{DefectArea}{BananaArea} \times 100$$

This percentage provides an objective measurement of banana quality.

Pseudocode:

Algorithm 8 Defect Detection

Input : Banana

Threshold Dark Regions

Morphology

Connected Components

Calculate Defect Area

Return Defect Percentage

3.3.9 Quality Grading and Shelf-Life Prediction

The final grading combines

- ripeness stage
- defect percentage

A rule-based decision engine assigns

- Grade A
- Grade B
- Grade C

The same information is used to estimate the remaining shelf life.

Example rules include:

- **Grade A:** Minimal defects with acceptable ripeness; suitable for normal sale.
- **Grade B:** Moderate defects or advanced ripeness; suitable for discounted sale or early consumption.
- **Grade C:** Severe defects or spoiled condition; recommended for processing or disposal if unsafe.

Shelf-life estimation follows the identified ripeness category, with unripe bananas having the longest remaining shelf life and spoiled bananas the shortest.

Pseudocode:

Algorithm 9 Quality Assessment

Input: Ripeness Stage, Defect Percentage

Assign Quality Grade

Estimate Shelf Life

Generate Recommendation

Return Results

3.3.10 Dashboard and Report Generation

After processing is complete, the SmartBanana system presents all analysis results through the dashboard. The dashboard displays the original image, enhanced image, segmentation mask, ripeness stage, defect percentage, quality grade, shelf-life estimation, and recommendations. Historical inspection records are stored to allow users to compare previous analyses over time.

The system also supports batch image processing, enabling multiple banana images to be analysed consecutively. For each processed image, a PDF report can be generated summarising all inspection results, including intermediate processing outputs and the final assessment. This reporting functionality provides users with a permanent record of the inspection process and supports quality monitoring in commercial or educational environments.

Pseudocode:

```
Algorithm 10 Report Generation

Input: Analysis Results



Display Dashboard

Store Record

Generate PDF Report

Return Report
```

Result & Discussion

4.1 Experimental Results

Instruction to Student: Demonstrate the definitive outcomes of your project based on your test plan. Use suitable data visualisations (e.g., bar charts, stacked line graphs, or pie charts) to illustrate performance or output metrics. Working prototypes as screenshots can be inserted but the results still be the focus.

4.2 Discussion and Critical Interpretation

Instruction to Student: Critically analyse your experimental results. Do not just state what the data is—interpret what the data means. Discuss the implications of your findings and provide a detailed comparison of how different algorithms or settings performed against each other.

Conclusion

5.1 Achievements

Instruction to Student: Synthesise your findings and explicitly state whether your project successfully fulfilled the SMART objectives outlined in Section 1.2. Declare which algorithm proved to be the best based on your experimental data.

- **Objective 1 Status:** [Fulfilled / Partially Fulfilled] – *Briefly explain how based on experimental data.*
- **Objective 2 Status:** [Fulfilled / Partially Fulfilled] – *Briefly explain how based on experimental data.*
- **Objective 3 Status:** [Fulfilled / Partially Fulfilled] – *Briefly explain how based on experimental data.*

Project Code Repository: *[Insert your GitHub Repository Link here, if applicable]*

5.2 Limitations and Future Works

Instruction to Student: Honestly discuss the current constraints, bugs, or boundaries of your prototype. Suggest concrete, actionable improvements or features that could be implemented in future iterations of the work.

References

Books

Gonzalez, R. C., & Woods, R. E. (2018). Digital image processing (4th ed.). Pearson.

Sonka, M., Hlavac, V., & Boyle, R. (2015). Image processing, analysis, and machine vision (4th ed.). Cengage Learning.

Journal Articles

Brosnan, T., & Sun, D.-W. (2004). Improving quality inspection of food products by computer vision—A review. Journal of Food Engineering, 61(1), 3–16. https://doi.org/10.1016/S0260-8774(03)00183-3

Chuquimarca, L., Salazar, A., Pérez, M., & Granda, E. (2024). A review of external quality inspection for fruit grading using CNN models. Smart Agricultural Technology, 8, 100501. https://doi.org/10.1016/j.atech.2024.100501

Chen, Y., Li, H., Zhang, X., et al. (2025). Real-time banana freshness grading: A portable end-to-end detection system with high precision. Computers and Electronics in Agriculture. https://doi.org/10.1016/j.compag.2025.110536

Official Organisations

Food and Agriculture Organization of the United Nations. (2023). FAOSTAT: Crops and livestock products. https://www.fao.org/faostat/

United Nations. (2015). Transforming our world: The 2030 Agenda for Sustainable Development. https://sdgs.un.org/2030agenda

Jabatan Penerangan Malaysia. (2023). Malaysia MADANI. https://malaysiamadani.gov.my/

Instruction to Student: Compile an exhaustive bibliography of all scholarly material referenced within this assignment. You must strictly adhere to the APA referencing format, ensuring each entry includes the author, publication year, title, and publisher or URL. Every in-text citation must have a matching full entry.

- *Specify the Digital Object Identifier (DOI) for your dataset where available.*
- *Avoid websites/forums/researchgate/arxiv if possible.*
- *Numbered lists are not permitted in this section.*
- *Organise all bibliographic entries in alphabetical order.*
- *Employ academic search engines or university library databases to ensure the precision and standardisation of your reference metadata.*

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Appendix A

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY
Group Contract Form

Faculty : Faculty of Computing and Information Technology (FOCS)
Programme :
Year of Study :
Semester :
Course Title : Image Processing
Course Code : BMDS2133
Group Assignment / Project Title :
Name of Tutor :
Tutorial Group :

Student Name and ID	Role(s)/Task(s) Assigned	Signature and Date

Appendix B

AI Usage Disclosure Form

Student Name : _____

Programme : _____

Course : _____

Assessment Title : _____

1. AI Tool(s) Used (Check that all that apply)

- ☐ None (I did not use any AI tools for this assignment)
- ☐ ChatGPT (Version: _____)
- ☐ Deepseek
- ☐ Gemini
- ☐ Other: _____

2. Nature of Assistance (Check all that apply)

- ☐ **Brainstorming:** Generating ideas, topics, or outlines.
- ☐ **Research:** Summarising long articles or finding concepts.
- ☐ **Editing:** Checking grammar, spelling, or sentence structure.
- ☐ **Coding/Math:** Debugging code or explaining a formula.
- ☐ **Creation:** Generating images, data, or draft text.

3. Disclosure Statement

Provide a brief explanation of how AI tools were utilised in completing the task and describe the measures taken to ensure that the final submission reflects your own independent work.

Example:
"I used ChatGPT to structure my initial ideas into an outline. Subsequently, I composed the full paragraphs independently and cross-checked the referenced dates against my textbook to ensure accuracy."

Your Statement:

4. Verification & Integrity

- ☐ I have reviewed and verified the accuracy of all facts, data, and citations generated with the assistance of AI tools.
- ☐ I have properly acknowledged and cited any AI-generated text, ideas, or content in accordance with my instructor's requirements.
- ☐ I confirm that the reasoning, interpretation, and analysis presented in the final submission are entirely my own.

Student Signature : _____

Date : _____

AMM 5.3.2026/SMC 1.4.2026